CSE 225: Data Structure and Algorithms
Midterm Exam, Fall 2020
North South University

Full Marks: 40 **Time: 40 minutes**

Student Name : ____

ID : _________________________________

Answering Guidelines

- Print this file and write your answer on the printed paper, scan (or take photos of) the answer script and submit the script in ONE PDF file.
- If you do not have access to a printer, then you can write your answers in white paper. You DO NOT need to write the questions.
- Try to keep your answer within the allotted space. If for some reason, you need more space, you can add extra pages. Clearly mention in the extra page which question you are answering.
- You may NOT consult textbooks or online sources.
- It is not allowed to consult with any person who has knowledge of this subject, including other students of this course. You may ask question to the instructor if you do not understand the question, but not more than that. All solutions have to be your own work.
- You must show all work for each problem to receive full credit.

CODE OF HONOR PLEDGE

I pledge on my honor that I have not given or received any unauthorized assistance on this exam.

Signature:____________________________

Date :____________________________

1. a) With the help of an example, discuss why we need a **reserved space** in the array implementation of a queue. 4

b) What are the advantages of arrays over linked lists? 3

c) What are the advantages of linked lists over arrays 3

2. For each of the following operations on data structures, mention the worst case running time in asymptotic (big-O) notation. **Justify your answer.** 10

a) Inserting an item in an unsorted list (array)

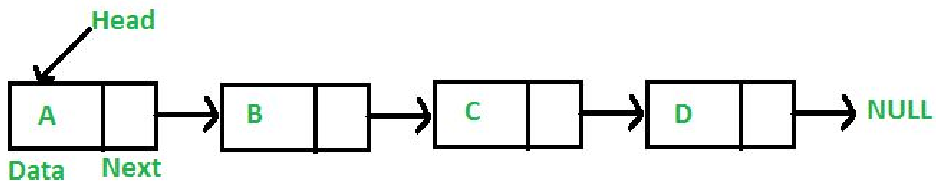
b) Deleting an item from a sorted list (array), where index is given

c) Searching an item in a sorted list (linked list)

d) Enqueue an item into a queue (linked list)

e) Inserting an item in a sorted list (linked list)

3. Suppose that the following linked list is given:



Write pieces of code to the following: {Don't need to write whole code, e.g., no main function needed]

a) Print 'C' [Hint: No loop needed] 2

b) Print all the letters 'A B C D' [You need a loop] 3

c) Insert a node with letter 'X' between 'B' and 'C'. [The first two lines to create a new node is written for you. You now have to write **three lines of code** to put the letter 'X' in the empty node, and make sure the node is placed between 'B' and 'C'. You don't need a loop.] 5

```
NodeType *newNode;
newNode = new NodeType;
```

4. a) We have the following C++ variable and pointer: 6

```
int num = 10;

int *ptr;

ptr = &num;
```

Suppose that the address of num is 2222 and the address of ptr is 5555. What will be the values for the following?

`num` ________________	`ptr` ____________________
`&num` ________________	`&ptr` ____________________
`*num` ________________	`*ptr` ____________________

b) How is a dangling pointer problem created? Show with an example. How can the problem be handled? 4